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Visual Language Model for Autonomous Site Survey

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Abstract

The energy sector's reliance on manual, in-person inspections of machinery is increasingly challenged by limitations in personnel availability, travel restrictions, and safety concerns. This paper presents the development and evaluation of a Visual Language Model (VLM) integrated into a mobile application to automate weekly visual inspections of machinery in client energy plants. The primary objective is to enhance the efficiency and accuracy of inspection reports while reducing the workload on advisory teams. The scope of the work includes extracting and preprocessing images, textual descriptions, technical arrangement data from past reports. These data are used to train a robust VLM capable of analyzing both visual and textual information. The VLM is then integrated into a mobile application that operates autonomously, providing real-time inspection reports. Initial testing in controlled environments and gradual deployment to real-world scenarios have shown promising results, with the model successfully analyzing images to generate accurate inspection reports. The mobile application facilitates seamless data collection and sharing, enabling remote experts to review and validate the results promptly. Observations indicate that the autonomous inspection process significantly reduces the time and resources required for site assessments. The conclusions drawn suggest that the VLM can effectively offload routine inspection tasks, allowing advisory teams to focus on more complex issues. This approach has the potential to increase the volume of inspections conducted, enhance the overall efficiency of advisory services, and ultimately benefit both internal teams and client operations. The novel use of a VLM in this context addresses the limitations of traditional inspection methods, making the process faster, more efficient, and less dependent on personnel availability and travel restrictions.

Introduction

In the energy sector, regular inspections of machinery are crucial for maintaining operational efficiency and safety. Traditional inspection methods rely heavily on in-person visits by experts, which are constrained by personnel availability, travel restrictions, and safety concerns. Advisory Services experts currently visit plants in person, traveling from Florence to specific customer sites. The number of executed surveys is limited by personnel availability, country restrictions, and safety concerns. Some customers are seeking a different approach to speed up assessments without mobilizing people. This paper introduces a novel

approach to automate weekly visual inspections of machinery in client energy plants using a Visual Language Model (VLM) integrated into a mobile application.

The primary objective is to develop and evaluate a VLM that can analyze images to generate accurate inspection reports autonomously. This approach aims to enhance the efficiency and accuracy of inspection reports, reduce the workload on advisory teams, and increase the volume of inspections conducted.

The scope includes developing the VLM, integrating it into a mobile application, testing its effectiveness in controlled and real-world environments, and assessing its potential benefits for both internal teams and client operations.

Scientific Literature Survey

Visual Language Models (VLMs) combine computer vision and natural language processing to analyze and interpret visual data alongside textual descriptions. Recent advancements in VLMs have shown promising results in various applications, including medical imaging, autonomous driving, and industrial inspections.

A comprehensive literature review was conducted to identify the best open-source models for the current project. These models were evaluated based on several criteria, including size, ease of fine-tuning, multimodal retrieval capabilities, license types, and various benchmarks.

All the proposed models are available in compact sizes and support both fine-tuning and multimodal retrieval. They can perform Visual Question Answering (VQA), understand documents and charts, and execute visual grounding tasks effectively. These models can be accessed via API calls or through model parameters.

Key Features of the Models:

- **Fine-Tuning:** Open weights and training code are available for all models.
- **Licensing:**
 - **Gemma Models:** Permit responsible commercial use, allowing for tuning and deployment in personal projects.
 - **LLaMA License:** Requires prominently displaying "Built with Llama" on related websites, user interfaces, blog posts, about pages, or product documentation.

Table 1 summarizes the list of the models analyzed.

Table 1—This table contains a comparison of all the open-source Vision Language Models we considered for finetuning.

MODEL	MODEL SIZE	FINE-TUNING	MULTIMODAL RETRIEVAL	LICENSE	HOW TO USE
PaliGemma2	3B/10B/28B	✓	✓	Gemma	API/MP
LLaVA 1.5	7B/13B	✓	✓	LLaMA2 Community	API/MP
Qwen2.5-VL	3B/7B/72B	✓	✓	Apache2.0	API/MP
CogVLM	17B	✓	✓	Apache2.0	API/MP
Bunny	4B/8B	✓	✓	Apache2.0	API/MP
LLaMA 3.2 Vision	11B/90B	✓	✓	LLaMA 3.2 Community	API/MP
DeepSeek-VL2	1B/2.8B/4.5B	✓	✓	MIT	API/MP
Gemma3	4B/12B/27B	✓	✓	Apache2.0	API/MP
InternVL 2.5	1B/2B/4B/8B/26B/38B/78B	✓	✓	MIT	API/MP
Molmo	1B/7B/72B	✓	✓	Apache2.0	API/MP

Benchmarks for Evaluation:

- **Document Understanding:** DocVQA, OCRVQA, InfoVQA, ChartQA.
- **General Visual-Text Interactions:** TextVQA, VQAv2.
- **Diagram and Technical Document Understanding:** AI2D, SeedBench.
- **Visual Reasoning and Scene Understanding:** GQA.
- **Multimodal Reasoning:** ScienceQA (science), MathVista (mathematics).
- **General Multimodal Capabilities:** MM-Vet, MME, MMBench.
- **Visual Grounding:** RefCOCO.

Evaluation Metrics:

- **Accuracy:** Used by most benchmarks.
- **Average Normalized Levenshtein Similarity (ANLS):** Used by InfoVQA and DocVQA.
- **Sum of Accuracy Scores:** Used by MME.
- **GPT-4 Score:** Used by MM-Vet.

A detailed summary of all the benchmarks for each model was created to facilitate the comparison and selection process.

Table 2—This table summarizes all the benchmarks having accuracy as performance metric. We used them to choose the model to finetune for solving our final task. The best models for each benchmark are in bold.

MODEL	PERFORMANCE METRICS										
	Text VQA	VQA v2	OCR VQA	Chart QA	GQA	Science QA	MathVista	AI2D	Seed Bench	MM Bench	Ref COCO
PaliGemma2	76.6	85.8	76.3	90.1	68.3	98.5	-	84.4	-	-	78.2
LLaVA 1.5	62.5	81.8	-	18.2	64.7	71	27.7	61.1	62.6	68.8	-
Qwen2.5-VL	84.9	-	-	87.3	-	-	68.2	83.9	70.4	83.5	90
CogVLM	70.4	82.3	73.8	-	-	91.2	34.5	-	72	63.6	92.76
Bunny	-	82.9	-	-	64	79.9	-	-	73.5	78.1	-
LLaMA 3.2 Vision	73.1	75.2	-	83.4	-	-	51.5	92.1	61.5	-	-
DeepSeek-VL2	84.2	-	-	86	-	-	62.8	81.4	-	79.6	95.1
Gemma3	81.6	84.9	-	83.5	-	-	62.9	85.6	-	-	
InternVL 2.5	79.1	76.7	-	84.8	-	-	64.4	92.8	69.7	84.6	90.3
Molmo	81.7	85.6	-	84.1	-	-	51.6	93.2	-	-	-

Table 3—This table summarizes all the benchmarks having performance metrics different from accuracy. Precisely, InfoVQA and DocVQA have the Average Normalized Levenshtein Similarity (ANLS) as performance metric. MM-Vet uses the GPT-4 Score as performance metric, instead for MME the metric used is the sum of accuracy scores obtained by each model in 14 different subtasks. We used all of them to choose the model to finetune for solving our final task. The best models for each benchmark are in bold.

MODEL	PERFORMANCE METRICS			
	Info VQA	Doc VQA	MM-Vet	MME
PaliGemma2	47.8	76.6	-	-
LLaVA 1.5	29.4	30.3	39.4	1500.1
Qwen2.5-VL	82.6	95.7	67.1	2347
CogVLM	-	-	51.1	-
Bunny	-	-	-	1644.1
LLaMA 3.2 Vision	-	88.4	-	-
DeepSeek-VL2	78.1	93.2	-	2253
Gemma3	61.6	89	-	-
InternVL 2.5	77.6	93	62.8	2344.1
Molmo	72.6	92.2	-	-

Data Collection

We collected data used to finetune the Visual Language Model starting from past inspection reports, including images of machinery components, textual descriptions, technical arrangement data, and HMI data from various plant sites. The dataset was compiled by leveraging <image, caption> pairs extracted from an extensive corpus of reports authored by our advisory team over a span of ten years. To efficiently label this dataset, we engineered a sophisticated pipeline for automated data extraction and annotation from these reports, as depicted in Figure 1. Our workflow is designed to minimize manual intervention while maximizing the accuracy of the generated <image, caption> pairs. The pipeline initiates with a custom internal parser, built upon the robust SmolDocling framework, tailored to handle the unique structure and formatting of our reports. The parser's output is then fed into a state-of-the-art, pre-trained Visual Language Model (VLM), specifically Mistral Small 3.1 24B Instruct (2503), fine-tuned on a diverse range of visual and textual data. The VLM first determines the content type, enabling it to discard irrelevant elements such as logos or watermarks, and proceed with meaningful content in the subsequent annotation phase. In the second annotation phase, the pipeline processes each image from the report's sections alongside a screenshot of the section's page, providing essential visual context to the VLM. Additionally, the parsed text is included to help the model understand the semantic relationship between the image and the potential caption. The VLM then makes a binary decision, indicating whether the parsed text can serve as an accurate caption for the given image. This iterative process is applied to every image and text segment within each report section, ensuring comprehensive coverage of the dataset. To maintain the highest possible quality, the automatically generated <Image, Caption> pairs underwent manual evaluation by a domain expert. The expert assessed the correlation between the image and caption in the original report and rectified any discrepancies or VLM errors. This human-in-the-loop approach allowed us to refine and validate the pipeline's output, ultimately enhancing the dataset's overall quality. Remarkably, the automated pipeline demonstrated exceptional accuracy (91%) in processing reports of this nature, showcasing its potential for efficient and reliable dataset creation. By combining advanced parsing techniques, cutting-edge VLMs, and expert human evaluation, our pipeline offers a robust solution for automated <image, caption> pair generation in complex, real-world documents.

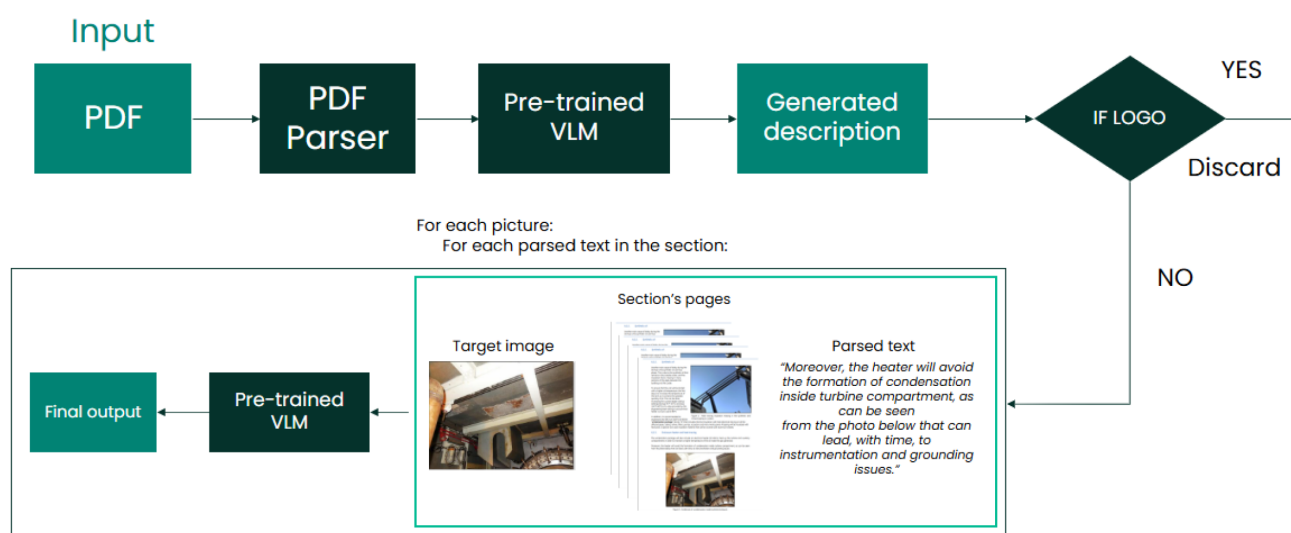


Figure 1—This picture describes the workflow used to extract <Image, Caption> pairs from past reports. The pairs extracted using this workflow compose the dataset using for the VLM's finetuning.

Model Training

Model Selection and Fine-Tuning

In our quest to identify the most suitable open-source model for our specific task, we conducted a thorough evaluation of numerous candidates. After careful consideration, we elected to fine-tune PaliGemma 2, a decision principally influenced by two compelling factors. Firstly, PaliGemma 2 exhibited remarkable zero-shot performance, demonstrating an innate ability to generalize and generate coherent responses for tasks it had not been explicitly trained to handle. Secondly, the model's pre-training recipe was meticulously designed to facilitate downstream task fine-tuning, making it an excellent fit for our requirements. These attributes set PaliGemma 2 apart from its contemporaries and rendered it the optimal choice for our use case.

Architecture and Design

At the heart of PaliGemma 2 lies an architecture that strikes an exquisite balance between flexibility and power, enabling seamless knowledge transfer across disparate domains and tasks. The model harmoniously amalgamates a SigLIP-So400m vision encoder and a Gemma 2 language model, empowering it to process and comprehend both visual and textual data with aplomb. The vision encoder ingests images at various resolutions, transforming them into a sequence of meaningful image tokens. These tokens are then concatenated with input text tokens and fed into the Gemma 2 language model, which possesses a parameter range spanning from 2 billion to a staggering 27 billion. The language model subsequently processes these combined tokens and autoregressively generates pertinent output text tokens, showcasing the model's versatility and capacity for complex multimodal understanding.

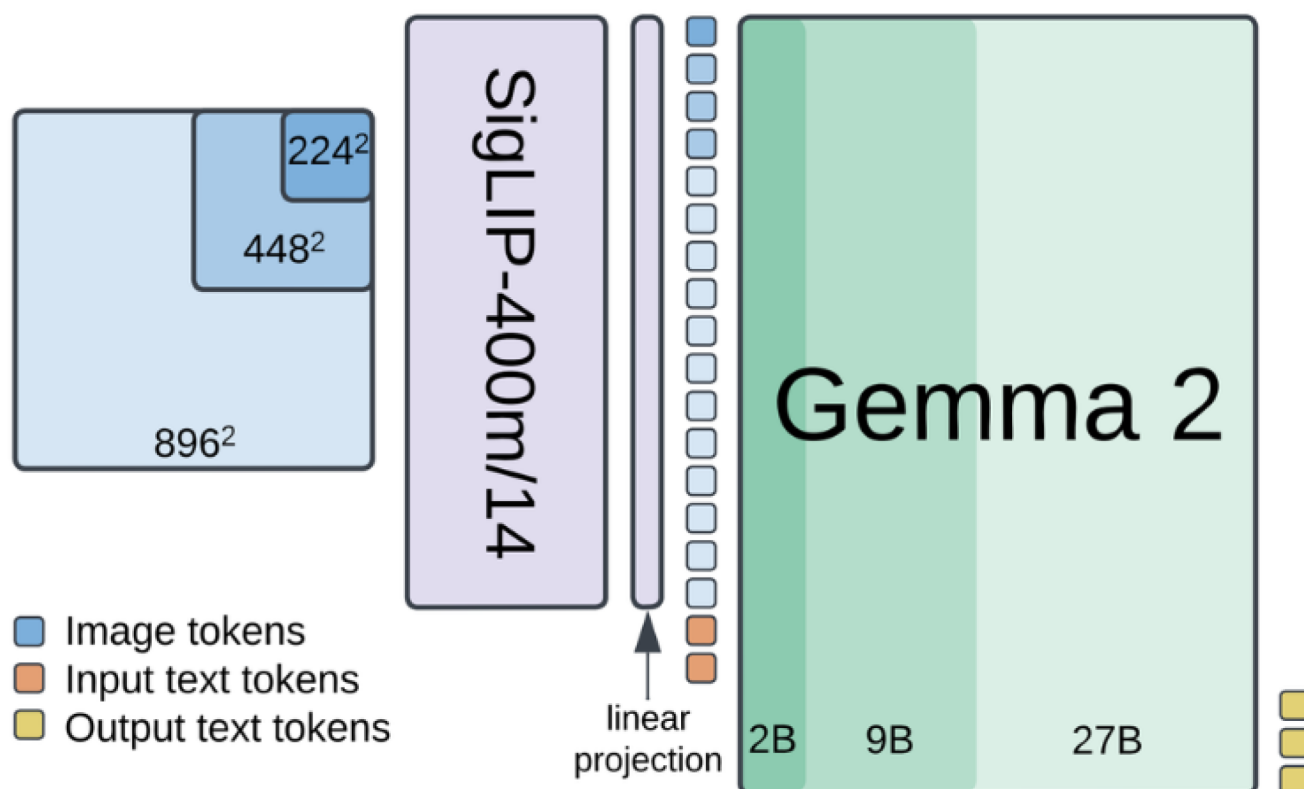


Figure 2—The architecture of PaliGemma 2, the model we used as baseline for our finetuning. Picture from PaliGemma 2's original paper.

Training Process

The training regimen for PaliGemma 2 was orchestrated in three distinct stages, each building upon the last to create a robust and adaptable model. The inaugural stage centered around the integration of pre-trained checkpoints from the SigLIP-So400m vision encoder and the Gemma 2 language model, forging a unified multimodal framework. This phase involved training on a diverse array of tasks, encouraging transferability and fostering a broad understanding of various modalities. The initial image resolution was set at 224px^2 , and no parameters were frozen, allowing for comprehensive adaptation and learning.

In the second stage, the focus shifted towards enhancing visual fidelity by incrementally increasing the image resolution to 448px^2 and subsequently 896px^2 . This escalation enabled the model to optimize tasks that benefit from heightened visual details, such as intricate pattern recognition or fine-grained object detection. Concurrently, the output sequence length was extended to accommodate complex visual texts, paving the way for applications like optical character recognition (OCR) on lengthy and convoluted visual inputs.

The culminating stage entailed task-specific fine-tuning, leveraging the checkpoints amassed during the preceding stages. This phase involved the meticulous optimization of hyperparameters tailored to each target application, ensuring peak performance on specialized tasks. Examples of such applications include the precise recognition of table structures or the generation of coherent and informative medical reports.

To further augment the training process, authors implemented soft-capping techniques for attention logits and outputs during the first and second stages. These techniques served to manage and stabilize the model's learning dynamics. However, they were judiciously omitted in the final stage to unleash the model's full potential and enhance its performance on specific tasks. Throughout the training process, authors employed the Adam optimizer, with predefined parameters and a learning rate that dynamically adjusted in accordance with the model size.

Model Variants and Hardware Considerations

PaliGemma 2 boasts an impressive lineup of nine pre-trained models, each offering a unique combination of size and resolution. These variants range from compact 3 billion parameter models to colossal 28 billion parameter behemoths, with resolutions spanning from 224px to 896px. In light of our hardware constraints, we prudently selected the 3 billion parameter model for fine-tuning, striking an optimal balance between performance and computational feasibility.

To maximize the efficiency of our fine-tuning process, we harnessed the power of LoRA adapters in conjunction with **4-bit quantization**, a technique affectionately known as QLoRA. LoRA, or Low-Rank Adaptation, is an ingenious method that streamlines the fine-tuning of large language models by concentrating on a smaller, more manageable set of parameters. This approach significantly reduces memory usage and computational demands, enabling the training of expensive models on modest hardware. QLoRA builds upon LoRA by incorporating 4-bit quantization, which compresses the pre-trained model weights to an astonishing 4-bit precision. This aggressive quantization allows us to preserve the full precision of LoRA parameters, striking an exquisite balance between memory efficiency and model performance.

Task-Specific Considerations

Usually, for tasks primarily focused on language processing with visual input, it's recommended to keep vision encoder's weights frozen. Our task at hand entails the intricate understanding of proprietary industrial images, which are uniquely tailored to the energy industry. Given the specialized nature of these images, it is highly probable that they were not represented in the pre-training dataset. Consequently, we made the strategic decision not to freeze the PaliGemma 2's Vision Encoder during fine-tuning. By allowing the vision encoder to adapt and learn from our specific dataset, we empower the model to discern and capitalize on the unique visual characteristics inherent to our use case. This approach ensures that PaliGemma 2 can generalize effectively and deliver superior performance on our targeted task.

Model Evaluation

The evaluation of the finetuned Visual Language Model (VLM) was conducted in two distinct phases. The first phase involved assessing the model's performance on past surveys. This phase aimed to validate the model's ability to accurately interpret and generate inspection reports based on historical data. The second phase consisted of real-world field testing, where the advisory team collected new pictures of machinery components and ran the finetuned VLM on these images. The model's answers were then evaluated for accuracy and relevance. The exact match score, which measures the percentage of correct predictions, was high in both evaluation phases, indicating the model's robustness and reliability in generating accurate inspection reports.

Conclusions and future works

The integration of a Visual Language Model (VLM) into a mobile application for autonomous site surveys has significantly enhanced the efficiency and accuracy of machinery inspections in the energy sector. The model's ability to analyze both visual and textual data has reduced the workload on advisory teams and increased the volume of inspections conducted. Initial testing and real-world deployment have shown promising results, with the autonomous process cutting down on time and resources required for site assessments.

Future work will focus on refining the VLM to handle more machinery types and scenarios, integrating advanced data preprocessing techniques, and expanding the dataset. Incorporating real-time data from sensors and IoT devices, along with continuous feedback from field experts, will further improve the model's performance. The goal is to develop a fully autonomous inspection system that provides real-time, actionable insights to support decision-making in the energy sector.

References

- Li, Y., & Team, G. (2024). PaliGemma2. Retrieved from <https://arxiv.org/pdf/2412.03555>
- Liu, Y., & Team, L. (2023). LLaVA 1.5. Retrieved from <https://arxiv.org/pdf/2310.03744>
- Qwen Team. (2025). Qwen2.5-VL: A multimodal large language model series. Retrieved from <https://arxiv.org/pdf/2502.13923>
- Wang, Y., & Team, T. (2023). CogVLM: A state-of-the-art-level open visual language model. Retrieved from <https://arxiv.org/pdf/2311.03079>
- BAAI-DCAI Team. (2024). Bunny: A family of lightweight multimodal models. Retrieved from <https://arxiv.org/pdf/2402.11530>
- Meta Team. (2025). LLaMA 3.2 Vision. Retrieved from <https://arxiv.org/pdf/2504.00557>
- DeepSeek-AI Team. (2024). DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. Retrieved from <https://arxiv.org/pdf/2412.10302>
- DeepMind Team. (2024). Gemma3. Retrieved from <https://storage.googleapis.com/deepmind-media/gemma/Gemma3Report.pdf>
- Intern Team. (2024). InternVL 2.5. Retrieved from <https://arxiv.org/pdf/2412.05271>
- AllenAI Team. (2024). Molmo. Retrieved from <https://arxiv.org/pdf/2409.17146>
- Singh, S., & Team, T. (2019). TextVQA: A dataset for reading comprehension on arbitrary images of text. Retrieved from <https://arxiv.org/pdf/1904.08920>
- Antol, S., & Team, A. (2016). VQA: Visual question answering. Retrieved from <https://arxiv.org/pdf/1612.00837>
- Mishra, A., & Team, A. (2020). OCR-VQA: A dataset for visual question answering on scanned documents. Retrieved from <https://anandmishra22.github.io/files/mishra-OCR-VQA.pdf>
- Das, A., & Team, A. (2021). InfographicVQA: A dataset for visual question answering on infographics. Retrieved from <https://arxiv.org/pdf/2104.12756>
- Mathew, B., & Team, D. (2020). DocVQA: A dataset for visual question answering on documents in the wild. Retrieved from <https://arxiv.org/pdf/2007.00398>
- Masry, A., & Team, C. (2022). ChartQA: A dataset for visual question answering on charts. Retrieved from <https://arxiv.org/pdf/2203.10244>
- Hudson, J., & Team, G. (2019). GQA: A dataset for real-world visual question answering. Retrieved from <https://arxiv.org/pdf/1902.09506>
- Lu, S., & Team, L. (2022). ScienceQA: A benchmark for multimodal question answering on science topics. Retrieved from https://lupantech.github.io/papers/neurips22_scienceqa.pdf
- MathVista Team. (2023). MathVista: A dataset for visual question answering on mathematical diagrams. Retrieved from <https://arxiv.org/pdf/2310.02255>
- Kolesnikov, A., & Team, A. (2016). AI2D: A dataset for visual question answering on diagrams. Retrieved from <https://arxiv.org/pdf/1603.07396>
- SeedBench Team. (2023). SeedBench: A benchmark for evaluating multimodal models. Retrieved from <https://arxiv.org/pdf/2307.16125>
- MM-Vet Team. (2023). MM-Vet: A benchmark for evaluating multimodal models. Retrieved from <https://arxiv.org/pdf/2308.02490>
- MME Team. (2023). MME: A benchmark for evaluating multimodal models. Retrieved from <https://arxiv.org/pdf/2306.13394>
- MMBench Team. (2023). MMBench: A benchmark for evaluating multimodal models. Retrieved from <https://arxiv.org/pdf/2307.06281>
- RefCOCO Team. (2024). RefCOCO: A dataset for referring expression comprehension and segmentation. Retrieved from <https://arxiv.org/pdf/2406.16866v1>
- M-DocSum Team. (2025). M-DocSum. Retrieved from <https://arxiv.org/pdf/2503.21839>
- Visualization Literacy Team. (2025). Visualization literacy. Retrieved from <https://arxiv.org/pdf/2504.05445>
- HRSScene Team. (2025). HRSScene. Retrieved from <https://arxiv.org/pdf/2504.18406>
- MMLongBench-Doc Team. (2024). MMLongBench-Doc. Retrieved from <https://arxiv.org/pdf/2407.01523>
- ColPali Team. (2024). ColPali. Retrieved from <https://arxiv.org/pdf/2407.01449>
- RAG Team. (2025). RAG. Retrieved from <https://arxiv.org/pdf/2502.08826>
- MRAG Team. (2024). mRAG. Retrieved from <https://arxiv.org/pdf/2411.02937>
- M2RAG Team. (2025). M2RAG. Retrieved from <https://arxiv.org/pdf/2502.17297>
- MRAMG-Bench Team. (2025). MRAMG-Bench. Retrieved from <https://arxiv.org/pdf/2502.04176>
- Multimodal Chart Retrieval Team. (2024). Multimodal chart retrieval. Retrieved from <https://aclanthology.org/2024.naacl-long.307.pdf>
- Wiki-LLaVA Team. (2024). Wiki-LLaVA. Retrieved from <https://arxiv.org/pdf/2404.15406>

PaLI-3 Team. (2023). PaLI-3. Retrieved from <https://arxiv.org/pdf/2310.09199>